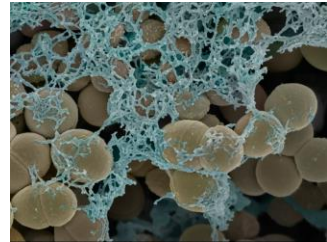


Periodontology 1

Host Response and Biofilm

ODON 621

Jennifer Goss, DDS
Assistant Professor



Mission: To advance the science and art of health through
education, service, scholarship and social accountability

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1



Learning Objectives

- Describe factors that contribute to susceptible host response
- Explain host's innate immune response to biofilm
- Explain contribution of genetics to host susceptibility
- Understand significance and formation of biofilm
- Know main groups of organisms found in health and disease
- Understand microbiome shifts that occur in disease
- Understand host response and biofilms affect therapeutic approaches

Classification at a Glance

Periodontal Health, Gingival Diseases and Conditions

- Periodontal Health and Gingival Health
- Gingivitis: Dental Biofilm-Induced
- Gingival Diseases: Non-Dental Biofilm-Induced

Periodontitis

- Necrotizing Periodontal Diseases
- Periodontitis
- Periodontitis as a Manifestation of Systemic Disease

Periodontal Manifestations of Systemic Diseases and Developmental and Acquired Conditions

- Systemic Diseases or Conditions Affecting Periodontal Supporting Tissues
- Mucogingival Deformities and Conditions
- Traumatic Occlusal Forces
- Tooth- and Prosthesis-Related Factors
- Periodontal Abscesses and Endodontic-Periodontal Lesions

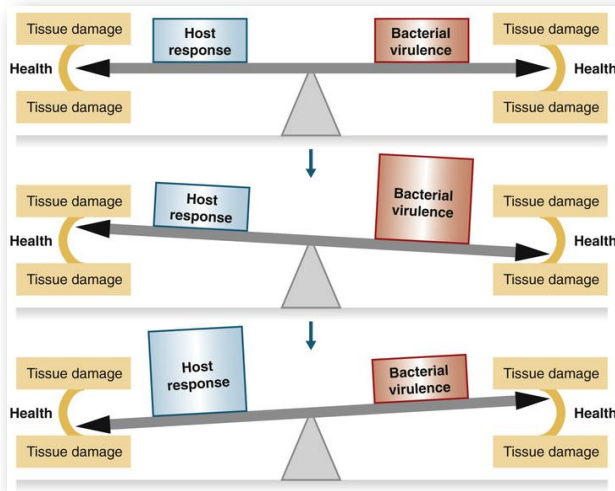
Peri-Implant Diseases and Conditions

- Peri-Implant Health
- Peri-Implant Mucositis
- Peri-Implantitis
- Peri-Implant Soft and Hard Tissue Deficiencies

2018 Classification of Periodontal and
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Gingivitis : Dental Biofilm induced, Periodontitis as a manifestation of systemic disease, Systemic diseases or conditions affecting periodontal supporting tissues will also be incorporated into today's discussion in addition to those from last week

Periodontitis



- Inflammatory disease initiated by bacteria
- Host response causes majority of damage to periodontium
- (damage = pathogenesis)

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4

Balance between bac flora and host response in health there is a balance between the 2 . There will always be bacteria present

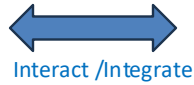
Host response is intended to be protective

Host response is strongly influenced by genetics

Host Response

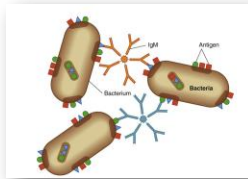
- Inflammatory Response (innate)

- Non-specific
- Epithelial barrier
- Antimicrobial molecules
- Complement cascade
- Cytokines
- Neutrophils
- Mast cells

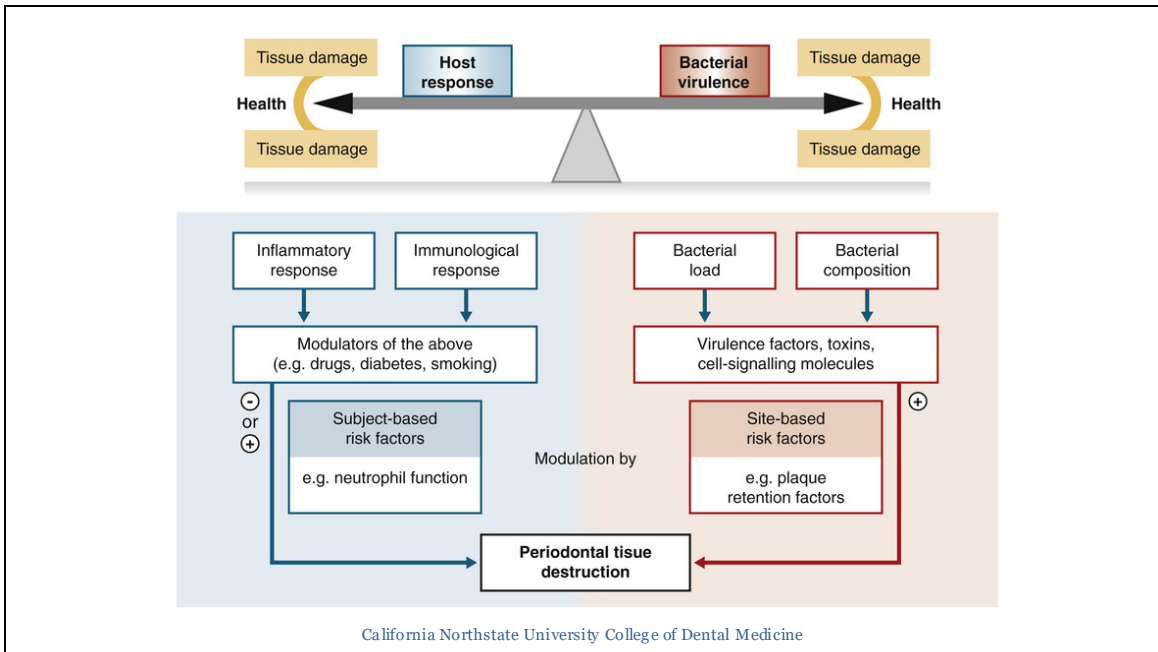


- Immune Response (adaptive)

- T-cells
- In periodontitis IgG (GCF) & IgA (Saliva) most important
- B-lymphocytes
- Antibodies/Immunoglobulins
 - Plasma cells
- In periodontitis IgG (GCF) & IgA (Saliva) most important

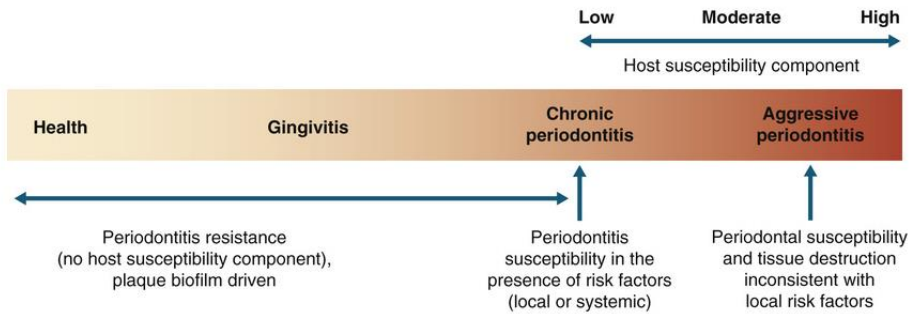


We will discuss Host response more next week



Subject based risk factors-we will go into detail on Risk Factors and Risk Assessment

Host Response



The more susceptible the host, the more damage will occur, termed “aggressive” meaning more tissue destruction than would be consistent with local risk factors

Gingival inflammation

- Gingival inflammation is regarded as a necessary prerequisite for the subsequent development of periodontitis and progressive attachment loss around teeth
- Management of gingivitis is therefore a key primary preventive strategy for periodontitis and a secondary preventive strategy for recurrence of periodontitis
- Management: Plaque control

“Normal” Host Response: Gingivitis

- Accumulation of biofilm
- Reversible inflammation
- No periodontal tissue or attachment loss
- Primarily innate immunity early lesions
- Chronic or progressive lesions:
 - more adaptive response



Photo: Karimbux, Nadeem, editor. Title: **Clinical cases in periodontics / edited by Nadeem Karimbux**. P. 11.

Description: Second edition. | Hoboken, NJ : Wiley-Blackwell, 2022. | Series: Clinical cases series |

Plaque induced gingivitis



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10

You will usually see plaque...but not always!

Probing depth in gingivitis



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11

Image: Periodontal-health.com

Plaque induced gingivitis

- Clinical signs : erythema, edema, bleeding on probing, tenderness, and enlargement (may be increased PD)
- Histopathologic changes:
 - elongation of rete ridges into the gingival connective tissue
 - vasculitis of blood vessels adjacent to the junctional epithelium
 - progressive destruction of the collagen fiber network with changes in collagen types
 - cytopathologic alterations of resident fibroblasts
 - progressive inflammatory/immune cellular infiltrate.

See article posted in Canvas from JP Murakami

Stages of Gingivitis Lesions

TABLE 7-1 Stages of Gingivitis

Stage	Time (Days)	Blood Vessels	Junctional and Sulcular Epithelia	Predominant Immune Cells	Collagen	Clinical Findings
I. Initial lesion	2–4	Vascular dilation Vasculitis	Infiltration by PMNs	PMNs	Perivascular loss	Gingival fluid flow
II. Early lesion	4–7	Vascular proliferation	Same as stage I Rete pegs Atrophic areas	Lymphocytes	Increased loss around infiltrate	Erythema Bleeding on probing
III. Established lesion	14–21	Same as stage II, plus blood stasis	Same as stage II but more advanced	Plasma cells	Continued loss	Changes in color, size, texture, etc.

PMNs, Polymorphonuclear leukocytes (neutrophils).

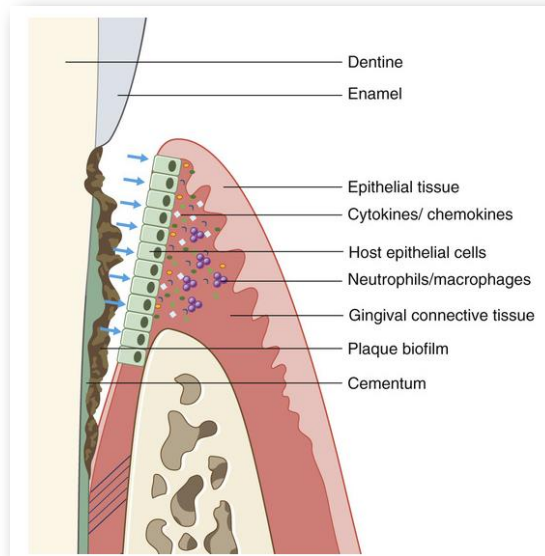
Carranza text

Cell signaling

Junctional epithelium
(JE)=host epithelial cells

Permeable to GCF &
PMNs

Cells recognize
MAMP/PAMPs
Intracellular signals
Cytokines released



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14

Bac from biofilm

Bac components such as LPS alert host presence of biofilm

Host releases PMS/neutrophils from gingival tissue and blood circulation first line defense

Promotion of host response via cell signaling via JE (junctional epith) and SE (sulcular epith)

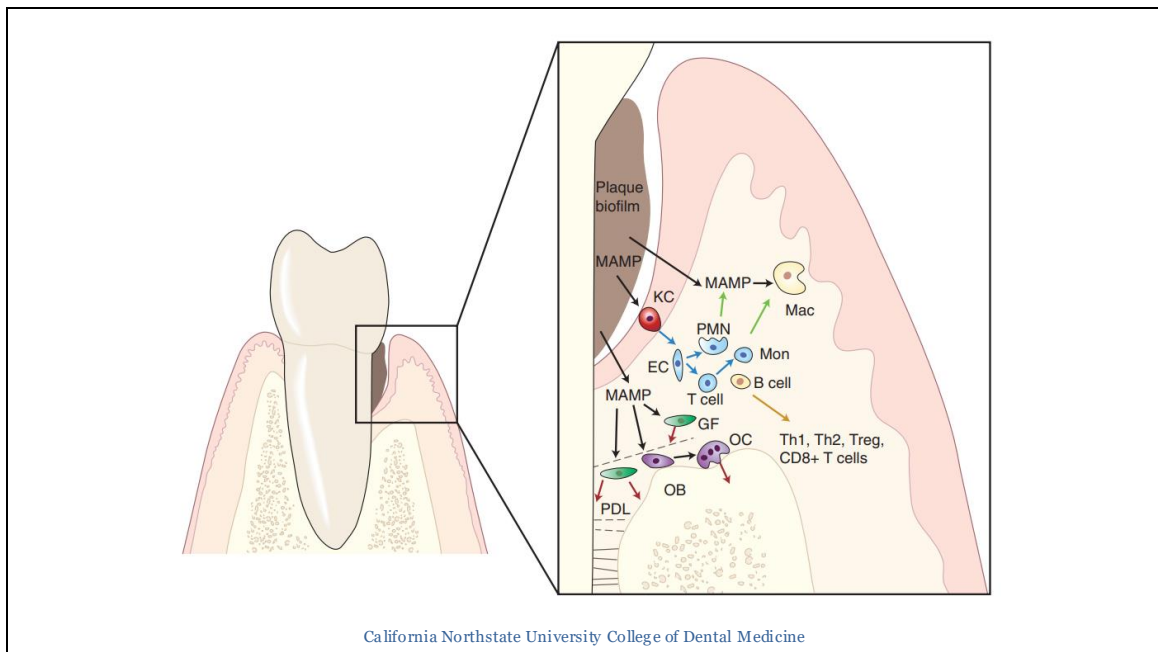
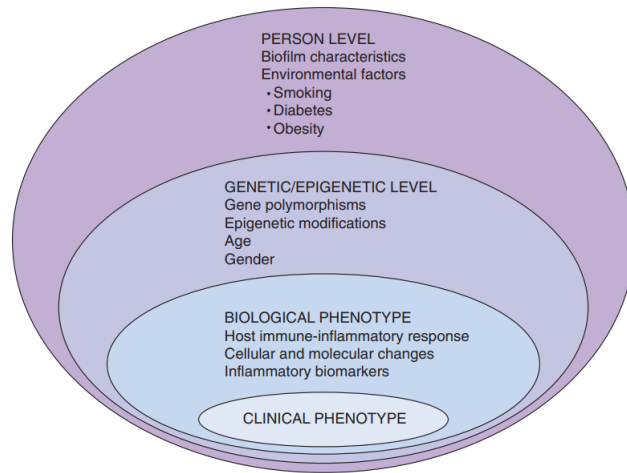


Figure 25-1 [Cell signaling] (Carranza text) Microbial-associated molecular pattern (MAMP) [**“AKA PAMPs”**] from microorganisms in the dental biofilm activate inflammatory responses in periodontal tissues. Biologic mediators will either affect neighboring cells (blue arrows) by inducing the expression of other mediators (e.g., RANKL) or by triggering chemotaxis (green arrows). Direct damage to periodontal tissues may also result after MAMPs stimulation (red arrows), such as metalloproteinase secretion by gingival and periodontal ligament fibroblasts. MAMP, Microbial-associated molecular patterns; RANKL, receptor activator of nuclear factor (NF) - κ B ligand; KC, keratinocyte; Mac, macrophage; GF, gingival fibroblast; PDL, periodontal ligament fibroblast; OB, osteoblast; OC, osteoclast; EC, endothelial cell; PMN, polymorphonuclear neutrophil; Mon, monocyte; T cell, T lymphocyte, B cell, B lymphocyte

Biologic systems model-Host response



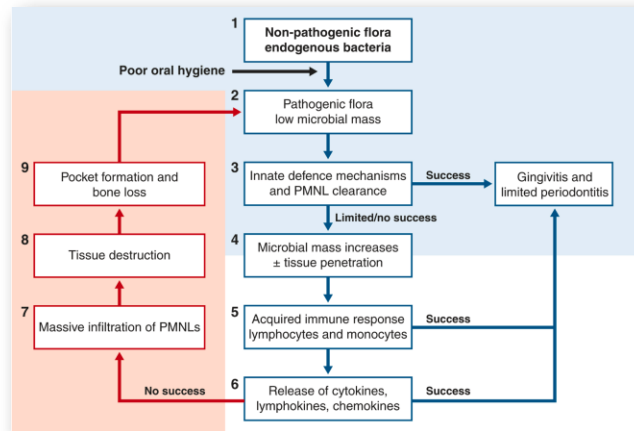
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16

Carranza: Figure 21-10 A biologic systems model for representing periodontitis. The outermost level of this model is the Person Level, representing an individual's unique characteristics as they impact on periodontitis. These include the compositional characteristics of the subgingival biofilm, and known risk factors and environmental exposures such as smoking and diabetes. The Person Level characteristics interact with the Genetic/Epigenetic Level characteristics, which include nonmodifiable factors such as age, gender, and genetic composition. Gene polymorphisms are known to be associated with periodontal disease, and epigenetics refers to changes in phenotype (i.e., clinical disease expression) caused by mechanisms other than changes in the underlying DNA sequence. Epigenetics can be defined as all the meiotically and mitotically inherited changes in gene expression that are not encoded in the DNA sequence itself, and epigenetic modifications are important permissive and suppressive factors in controlling the expressed genome via gene transcription. Two major epigenetic mechanisms are posttranslational modification of histone proteins in chromatin and methylation of DNA. The Genetic/Epigenetic Level characteristics influence the Biological Phenotype, which is characterized by the specific immune-inflammatory responses (cellular and molecular events and production of inflammatory mediators) that are associated with the Clinical Phenotype (i.e., the clinical presentation of the disease). This model reflects how

different individuals with the same presentation (e.g., periodontitis) may have very different predisposing and risk factors. The model depicts the different biologic factors that underpin the development of periodontal disease in different individuals and ultimately may be used to classify disease according to the contribution provided to the clinical phenotype at each level. (Modified from Offenbacher S, Barros SP, Beck JD: J Periodontol 79:1577-1584, 2008.

Critical Pathway Model

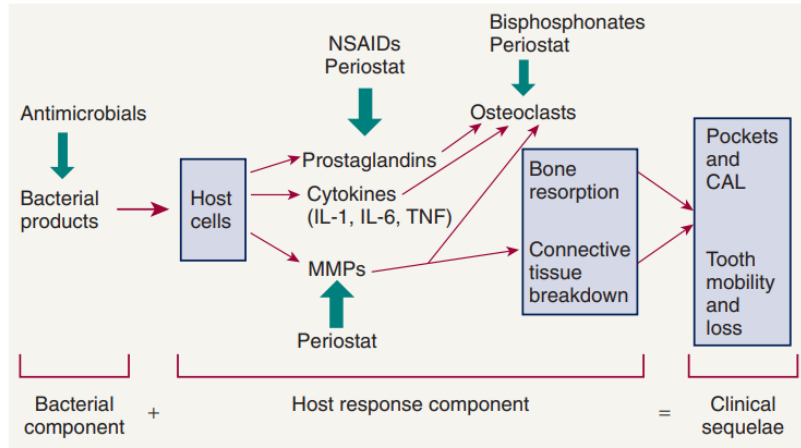


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17

Help understand the role of systemic risk factors in pathogenesis, stages that must be present for the disease to develop

Gene polymorphisms: overproduction IL-1



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18

Treatment potential: such as Periostat (doxycycline) blocks mediators

Inherited conditions: Additional risk development of periodontitis

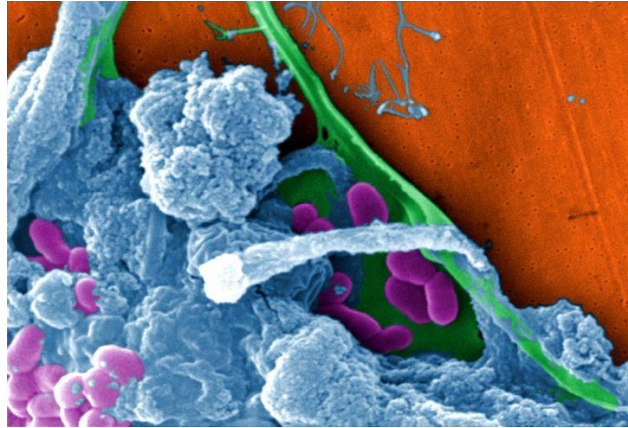
Condition	Associated gene	Underlying defect of periodontal relevance
Down's syndrome	Chromosome 21	Defects of neutrophil chemotaxis, killing and phagocytosis; depressed T cell antigen induced killing
Chronic granulomatous disease	Gp91 protein p91-PHOX on chromosome b558	Failure of respiratory burst in phagocytosis, oxygen radicals not produced & bacteria survive
Type 1 diabetes (insulin-dependent)	IDDM1 gene located on chromosome 6	Hyperglycemia reduces neutrophil function, Monocytes are hyperreactive excess IL-1, PGE-2, TNF- α . Collagen solubility, vascularity
Papillon-Lefevre syndrome	Cathepsin C gene on chromosome 11	Defects of neutrophil chemotaxis & phagocytosis
Ehlers-Danlos syndrome	Gene mutation for fibrous proteins/enzymes	Altered collagen structure and function, Type VII is associated with periodontal destruction
Chediak-Higashi syndrome	Lysosomal trafficking regulator gene	Defects of phagocytosis chemotaxis, degranulation: loss of adult dentition
Job syndrome	STAT3 gene on chromosome 17	Excess IgE & histamine; abn neutrophil chemo

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19

Cathepsins are the most abundant lysosomal proteases, mainly found in acidic endo/lysosomal compartments where they play a vital role in intracellular protein degradation, energy metabolism, and immune responses among a host of other functions

Biofilm



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20

Biofilm

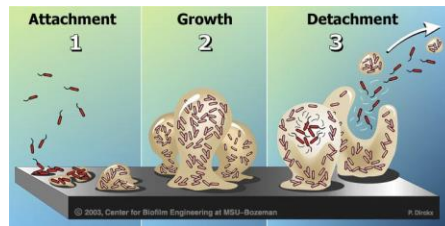
- Spatially and functionally organized multispecies microbial community
- Heterogeneous, organisms coexist (although they may appear incompatible)
- More tolerant of environmental stresses, host defenses, more difficult to treat
- Ubiquitous (everywhere)

Biofilms in the mouth

- Found on all surfaces (mucosal and dental)
- Dental plaque most diverse
- Resident oral microbiome confers benefits
 - Resist colonization by exogenous species
 - Down regulate host response **
 - Regulate GI and CV systems

Biofilm characteristics

The bacteria associated with periodontal disease reside within **biofilms** above and below the gingival margin

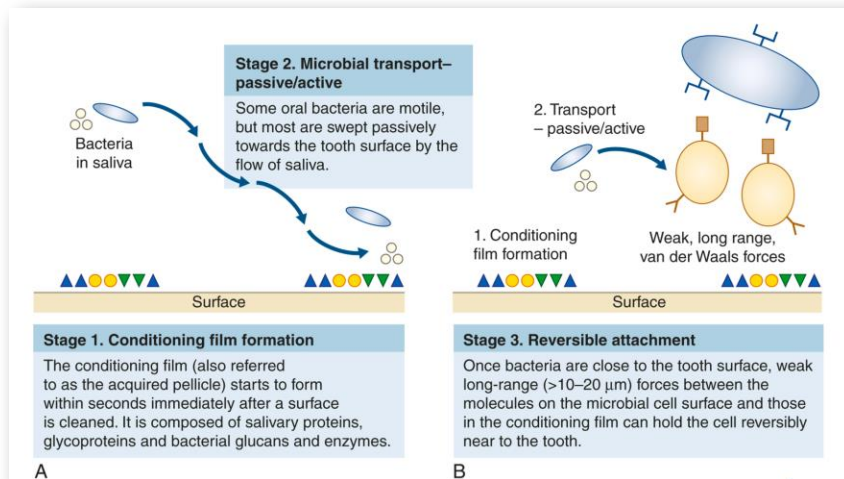


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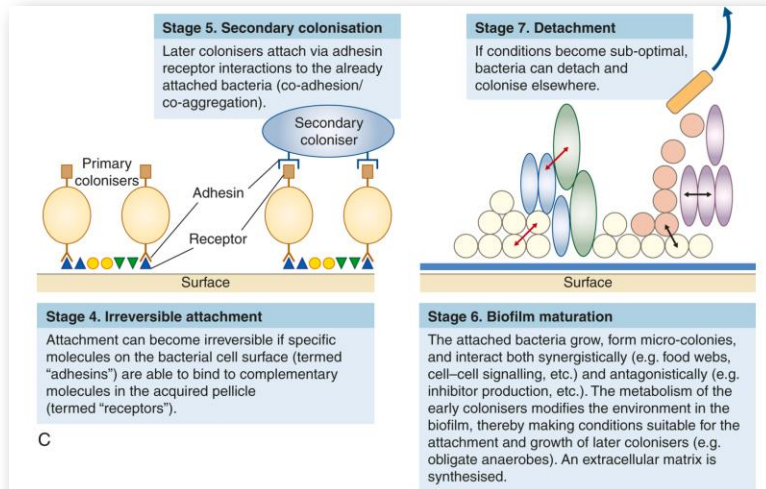
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Up to 1 billion bacteria may be present in the pocket, and pockets range from 4 to 12 millimeters in depth.¹ After they proliferate, some of the bacteria can detach from their colony and recognize elsewhere. Your text goes into more detail of the process of biofilm development-important because this can impact how we treat and manage disease

Development of Biofilm

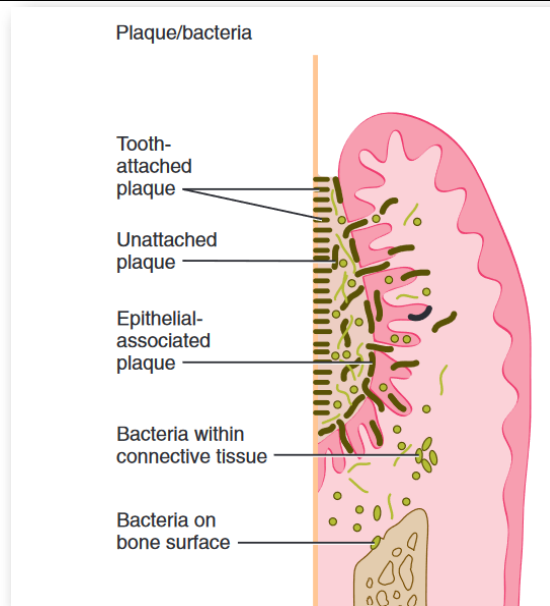


Development of Biofilm



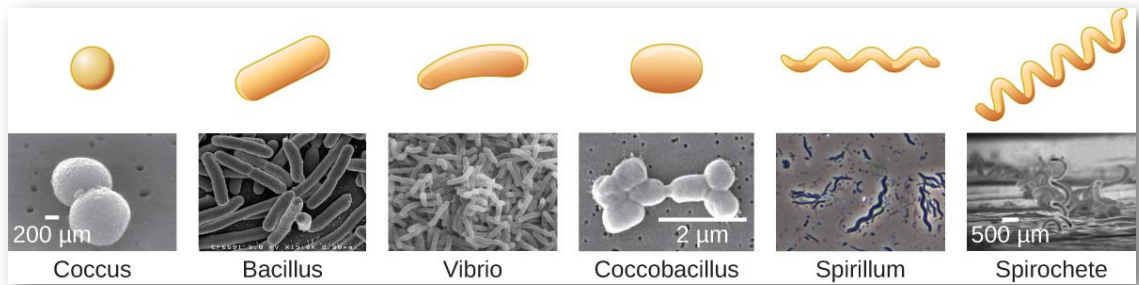
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25



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Bacteria are everywhere and vary in different ecosystems



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Bacteria shapes

Periodontal Disease Organisms



- Gram negative anaerobes
- Mixed infection
- Disease entities have patterns

TABLE 23-12 Summary of Bacterial Species Significantly Associated with Different Clinical Conditions (Based on Culture, Polymerase Chain Reaction (PCR), Checkerboard Data)

Health	Gingivitis	Chronic Periodontitis	Localized Aggressive Periodontitis	Aggressive Periodontitis	Necrotizing Periodontal Diseases	Abscesses of the Periodontium	Periimplantitis
<i>Actinomyces gerencseriae</i>	<i>Actinomyces naeslundii</i> 1	<i>Aggregatibacter actinomycetemcomitans</i> (serotype b)	<i>Capnocytophaga</i> spp.	<i>Aggregatibacter actinomycetemcomitans</i> (serotype b)	<i>Fusobacterium nucleatum</i>	<i>Fusobacterium nucleatum</i>	<i>Capnocytophaga</i> spp
<i>Actinomyces israelii</i>	<i>Actinomyces viscosus</i>	<i>Campylobacter rectus</i>	<i>Aggregatibacter actinomycetemcomitans</i> high (Ltx clone)	<i>Eubacterium nodatum</i>	<i>Prevotella intermedia</i>	<i>Parvimonas micra</i>	<i>Aggregatibacter actinomycetemcomitans</i>
<i>Actinomyces naeslundii</i> 1	<i>Streptococcus mitis</i>	<i>Eikenella corrodens</i>	<i>Campylobacter rectus</i>	<i>Prevotella intermedia</i>	<i>Spirochetes</i>	<i>Prevotella intermedia</i>	<i>Campylobacter rectus</i>
<i>Actinomyces oris</i>	<i>Streptococcus oralis</i>	<i>Eubacterium nodatum</i>	<i>Eikenella corrodens</i>	<i>Spirochetes</i>	<i>Treponema</i> spp.	<i>Parvimonas micra</i>	<i>Fusobacterium nucleatum</i>
<i>Actinomyces viscosus</i>	<i>Streptococcus sanguinis</i>	<i>Fusobacterium nucleatum</i>	<i>Eubacterium nodatum</i>	<i>Treponema</i> spp.	<i>Porphyromonas gingivalis</i>	<i>Spirochetes</i>	<i>Prevotella intermedia</i>
<i>Capnocytophaga gingivalis</i>	<i>Veillonella parvula</i>	<i>Parvimonas micra</i>	<i>Fusobacterium nucleatum</i>	<i>Tannerella forsythia</i>	<i>Treponema denticola</i>	<i>Porphyromonas gingivalis</i>	<i>Spirochetes</i>
<i>Capnocytophaga ochracea</i>	<i>Aggregatibacter actinomycetemcomitans</i> serotype a	<i>Prevotella intermedia</i>	<i>Prevotella intermedia</i>	<i>Campylobacter gracilis</i>	<i>Campylobacter showae</i>		<i>Treponema</i> spp
<i>Capnocytophaga</i> spp.	<i>Campylobacter concisus</i>	<i>Spirochetes</i>	<i>Spirochetes</i>	<i>Treponema denticola</i>	<i>Prevotella nigrescens</i>		<i>Enterobacteriaceae</i> sp
<i>Capnocytophaga sputigena</i>	<i>Campylobacter rectus</i>	<i>Treponema</i> spp.	<i>Treponema</i> spp.	<i>Campylobacter gracilis</i>			<i>Porphyromonas gingivalis</i>
<i>Eubacterium saburreum</i>	<i>Leptotrichia buccalis</i>	<i>Enterobacteriaceae</i> spp.	<i>Leptotrichia buccalis</i>	<i>Campylobacter showae</i>			<i>Tannerella forsythia</i>
<i>Fusobacterium periodonticum</i>	<i>Eikenella corrodens</i>	<i>Porphyromonas gingivalis</i>	<i>Porphyromonas gingivalis</i>				<i>Treponema denticola</i>
<i>Fusobacterium polymorphum</i>	<i>Eubacterium nodatum</i>	<i>Selenomonas noxia</i>	<i>Tannerella forsythia</i>				<i>Pseudomonas aeruginosa</i>
<i>Neisseria</i> spp.	<i>Fusobacterium nucleatum</i>	<i>Tannerella forsythia</i>	<i>Treponema denticola</i>				<i>Staphylococcus aureus</i>
<i>Propionibacterium acnes</i>	<i>Haemophilus</i> spp.	<i>Prevotella nigrescens</i>					<i>epidermidis</i>
<i>Streptococcus anginosus</i>	<i>Parvimonas micra</i>						

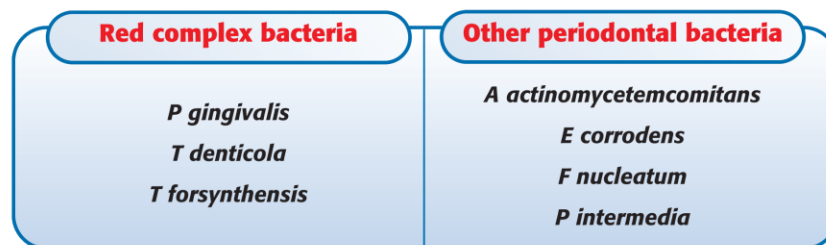
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Bacterial species associated with different clinical conditions

Bacteria implicated in Periodontal disease

There is a direct association between red complex bacteria and 2 of the most meaningful parameters in periodontal disease diagnosis¹:

- Pocket depth
- Bleeding on probing



References: 1. Socransky SS, Haffajee AD, Cugini MA, Smith C, Kent RL Jr. Microbial complexes in subgingival plaque. *J Clin Periodontol* 1998;25:134-144. 2. Socransky SS, Haffajee AD. Dental biofilms: difficult therapeutic targets. *Periodontol* 2000 2002;28:12-55.

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Dr. Sigmund Socransky and Dr. Anne Haffajee, periodontal thought leaders, have identified a specific group of bacterial pathogens linked to periodontal infection. This group of bacteria is known as the **red complex**, a name assigned as part of the doctors' color-coded system for grouping bacterial pathogens that tend to occur together in dental plaque. Late colonizers (BACTEROIDES FORSYTHUS) G-NONMOTILE orange. Early colonizers (green)

Red complex bacteria are commonly found in plaque when a periodontal infection occurs, and this persistent group of pathogens includes *Porphyromonas gingivalis*, *Treponema denticola*, and *Tannerella forsythensis*.

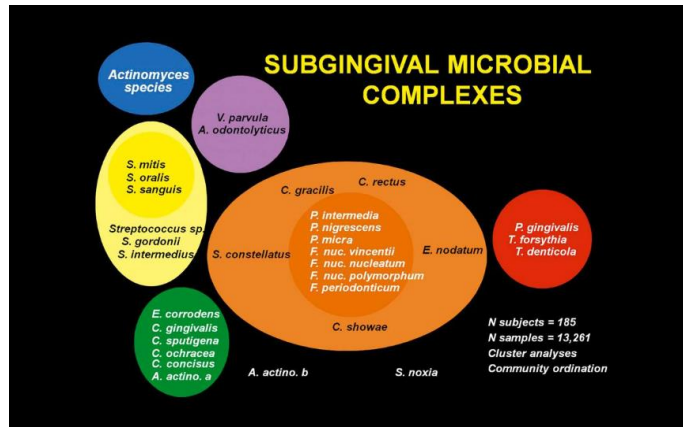
In studying the red complex, Drs. Socransky and Haffajee arrived at a very important finding. There is a direct association between large numbers of red complex bacteria and increased pocket depth and bleeding on probing—2 clinical parameters important to the diagnosis of periodontal infection.¹

Drs. Socransky and Haffajee also found that in addition to red complex bacteria, other bacterial pathogens appear to be associated with the status and progression of periodontal disease.¹

For example:

- *A actinomycetemcomitans* and *E corrodens* are early colonizers of the tooth surface^{1,2}
- *F nucleatum* are gram-negative pathogens that become dominant in the plaque layers at a later stage²
- *P intermedia* are spirochetes that have been implicated in causing periodontal infection. This species possesses “fibrils” (hair-like structures) that aid in the attachment to oral surfaces²

Red Complex Bacteria



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31

Socransky and Haffajee

TABLE 23-10 Prevalence of Microbial Species Associated with Various Clinical Forms of Periodontitis

Species	FORMS OF PERIODONTITIS		
	Chronic Periodontitis	Localized Aggressive Periodontitis	Aggressive Periodontitis
<i>Aggregatibacter actinomycetemcomitans</i>	28.2%	81.8%	40.9%
<i>Porphyromonas gingivalis</i>	53.8%	13.3%	79.6%
<i>Prevotella intermedia/nigrescens</i>	50.2%	53.4%	71.4%
<i>Tannerella forsythia</i>	50.6%	0	50.8%
<i>Campylobacter rectus</i>	40.3%	12.5%	47.8%

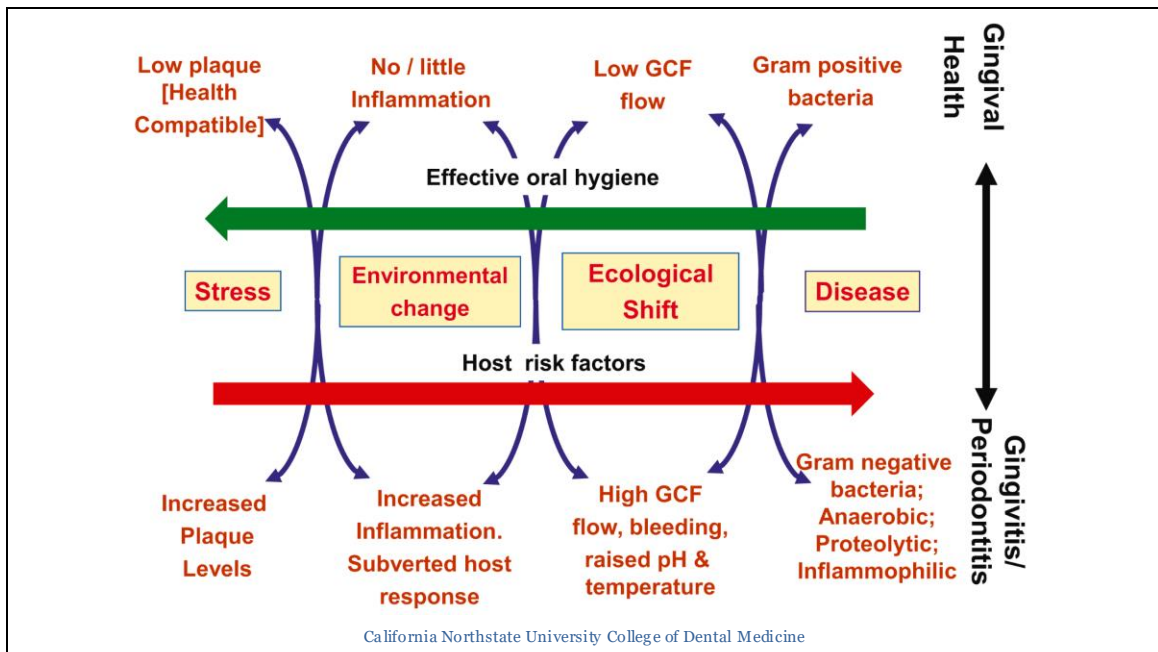
Adapted from Haffajee AD, Socransky SS: *Periodontol* 2000 5:78, 1994; and Mombelli A, Casagni F, Madianos PN: *J Clin Periodontol* 29 Suppl 3:10, 2002.
O, Occasionally isolated.

TABLE 23-11 Prevalence of Key Pathogens in Healthy Subjects and Patients with Periodontitis

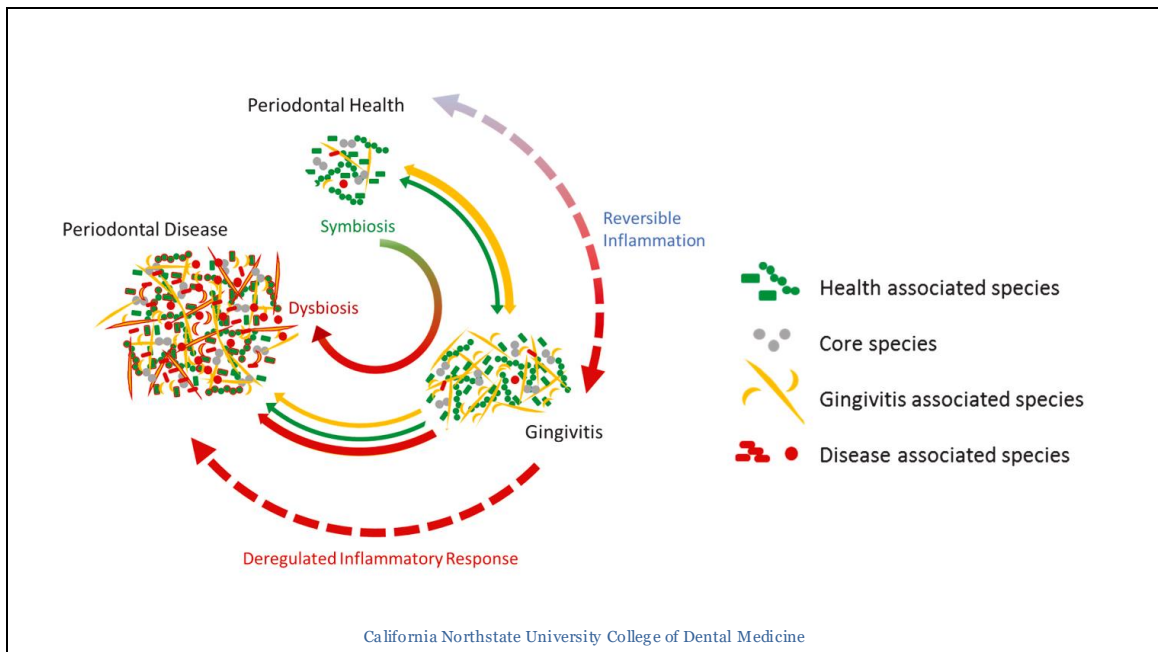
Species	PREVALENCE		SIGNIFICANCE OF DIFFERENCE	
	Health	Periodontitis	P-Value	Odds Ratio*
<i>Aggregatibacter actinomycetemcomitans</i>	12.8%	31.0%	0.002	3.1
<i>Porphyromonas gingivalis</i>	10.6%	59.5%	<0.001	12.3
<i>Prevotella intermedia/nigrescens</i>	69.1%	87.9%	0.001	3.3
<i>Tannerella forsythia</i>	47.9%	90.5%	<0.001	10.4
<i>Fusobacterium nucleatum</i>	85.1%	95.7%	0.014	3.9

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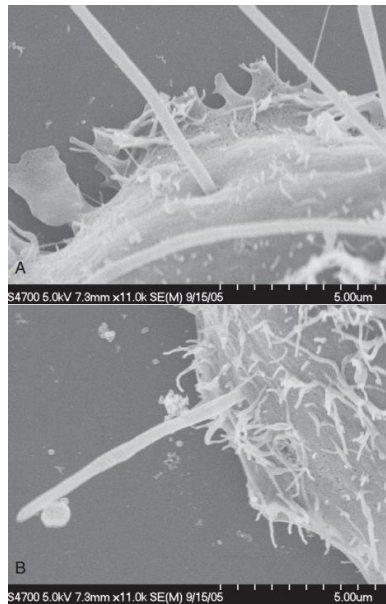
Carranza



Marsh 2003 ecological plaque hypothesis-a cyclical situation develops where conditions are created by host response to initial biofilm that selects for pathogens which continue to drive the inflammatory response



Bidirectional relationship between subg microbiome and inflammatory and immune response

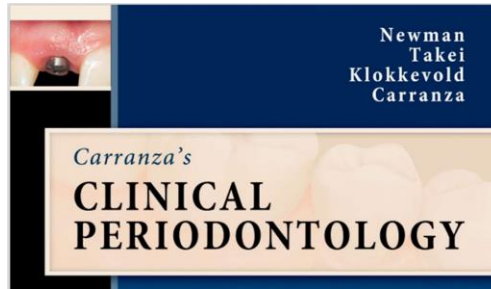
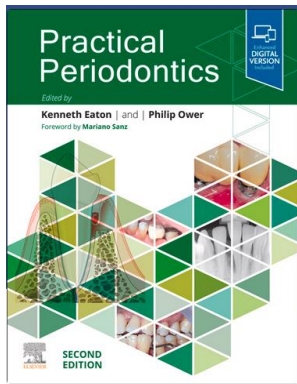


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Bacteria (*Fusobacterium nucleatum*) invading an epith cell from Carranza text



Resources



For extra reading, please refer to the above-mentioned resources.